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ORIGINAL PAPER

Parametric quaternion Wigner–Ville distribution: definition, uncertainty principles, and application

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Abstract
The Wigner–Ville distribution (WD) is a fundamental tool in signal processing and time–frequency analysis. Parametric embedding provides a practical approach to enhancing the effectiveness and flexibility of WD, expanding its utility in complex signal analysis. This paper presents a novel extension of the WD into the quaternion domain through parametric embedding, resulting in the parametric quaternion Wigner–Ville distribution (PQWD). We begin by presenting the formal definition of the PQWD and proceed to explore its key properties. Furthermore, we derive the uncertainty principles associated with the PQWD, highlighting its significant implications in mathematics, signal processing, and physics. Finally, we discuss the potential applications of PQWD in feature extraction, highlighting its versatility and potential impact in advanced signal analysis.

Keywords Linear canonical transform · Quaternion Wigner–Ville distribution · Quaternion linear canonical transform · Uncertainty principle

1 Introduction

The Wigner–Ville distribution (WD) and ambiguity function (AF) are powerful tools for time-frequency analysis and non-stationary signal processing in the complex domain. The WD captures energy distribution in the time-frequency domain, while the AF reveals autocorrelation in the time-delay and frequency-shift domain [1, 2]. Despite its strengths, the WD encounters bottlenecks when processing complex signals and faces challenges related to cross-term interference. To overcome these issues, researchers have improved the performance and adaptability of the WD through parametric embedding, often utilizing the linear canonical transform (LCT) [3–6].

The LCT is a three-parameter integral transform, generalizing the Fourier transform (FT), the fractional Fourier transform (FRFT), and the Fresnel transform (FST) [7, 8]. Combining LCT with WD and AF has led to the development of new signal processing tools that are more flexible and effective due to the additional degrees of freedom [9]. The LCT is introduced into WD and AF by replacing the Fourier transform kernel with the non-orthogonal kernels of LCT, resulting in the development of WDL and AFL, respectively [10]. These modifications provide enhanced flexibility and performance [10, 11]. A new type of WD based on LCT, termed UWA, deforms the instantaneous autocorrelation function (IAF) and generalizes both WDL and AFL, with studies focusing on its special cases [1, 3, 4, 12]. Further advancements include a scaled WD, parameterized by a scale factor k, which improves cross-term suppression [5]. The τ-WD framework also expands, allowing for a single scale change, $\tau \in [0, 1]$, and finds applications in time-frequency analysis, signal processing, and quantum mechanics [13–15].

Hypercomplex numbers, including quaternions, provide a broader algebraic framework than complex numbers and have shown superior performance in applications such as color and multispectral image processing, filtering, watermarking, pattern recognition, and edge detection [16, 17]. As a fourth-order Cayley–Dickson algebra, quaternions are widely utilized in signal processing and optics. While signal theory in the complex domain is relatively mature, quater-

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nion signal processing remains a challenging and evolving field. Building on the robust foundation of complex-domain transformations, quaternion-domain transformations have emerged as a key area of research, presenting numerous open problems. Recently, integral transforms, such as the quaternion Fourier transform (QFT), quaternion linear canonical transform (QLCT), and quaternion Wigner–Ville distribution (QWD), have been extended into the quaternion domain [18–20]. These quaternion-domain transforms are increasingly applied in fields including image processing, information security, and pattern recognition [21]. The QWD, which extends the WD into the quaternion domain, is particularly notable for its association with the LCT, an area of growing research interest [2, 22, 23]. Despite recent progress, significant opportunities remain to further develop and optimize QWDs, making it an active and promising direction in quaternion signal processing.

This paper introduces a novel definition of the WD in the quaternion domain through parametric embedding, resulting in the parametric quaternion Wigner–Ville distribution (PQWD), and investigates its key properties. This new formulation enhances flexibility and demonstrates significant potential for applications in signal processing. The uncertainty principle, which establishes a lower bound for the product of a signal and its Fourier transform [24], plays a crucial role in mathematics, signal processing, and physics. Substantial research has been conducted on the uncertainty principle for the LCT, WD, and their related variants [14, 15, 25]. Recent studies have extended this principle to the QLCT, defining the lower bound for the product of the effective widths of quaternion signals in the spatial and frequency domains [26–28]. Moreover, various formulations of the QLCT have been proposed and further refined [28, 29]. Building on this foundation, this paper will further explore the uncertainty principles associated with the newly defined PQWD, aiming to provide deeper insights into its theoretical framework and practical implications. The main contributions of this paper are as follows:

- The definition of the PQWD is presented.
- The key properties of the PQWD are analyzed.
- The uncertainty principles of the PQWD are derived.
- The application of the PQWD in feature extraction is explored.

The structure of this paper is organized as follows. Section 2 provides a brief overview of the fundamental concepts in quaternion algebra, the QLCT, and the QWD. In Sect. 3, we introduce the definition of the PQWD and explore its key properties. Section 4 presents the derivation of the uncertainty principles. In Sect. 5, we discuss potential applications of PQWD in feature extraction. Finally, Sect. 6 concludes the paper with a discussion of future research directions. In the discussion noted down, the square of a vector is defined as $\mathbf{a}^2 = (a_1^2, a_2^2)$, where $\mathbf{a} = (a_1, a_2)$. The inner product and element-wise division between vectors are expressed as $\mathbf{a} \cdot \mathbf{b} = (a_1 b_1, a_2 b_2)$ and $\mathbf{a}/\mathbf{b} = (a_1/b_1, a_2/b_2)$, where $\mathbf{b} = (b_1, b_2)$.

2 Preliminaries

2.1 The quaternion algebra

Quaternions, one of the hypercomplex numbers, are normed and noncommutative division algebras in which every non-zero element has an inverse.

Definition 1 The set of quaternions is denoted by $\mathbb{H}$. Each element of $\mathbb{H}$ can be expressed in the form of a Cartesian product [26]

$$\begin{aligned}\mathbb{H} =&\{q \mid q = [q]_0 + \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3; [q]_0, [q]_1,\\ &[q]_2, [q]_3 \in \mathbb{R}\},\end{aligned} \tag{1}$$

where the units $\mathbf{i}, \mathbf{j}, \mathbf{k}$ follow Hamilton's multiplication rules $\mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \mathbf{ki} = -\mathbf{ik} = \mathbf{j}, \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \mathbf{i^2} = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1$.

For a quaternion $q = [q]_0 + \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3 \in \mathbb{H}$, let $Sc(q) = [q]_0$ and $Ve(q) = \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3$ denote the scalar part and the vector part of q, respectively. That is, the quaternion $q \in \mathbb{H}$ can be expressed as $q = Sc(q) + Ve(q)$, and q is called pure quaternion when $Sc(q) = [q]_0 = 0$. The conjugate of a quaternion q is given by $q^* = [q]_0 - \mathbf{i}[q]_1 - \mathbf{j}[q]_2 - \mathbf{k}[q]_3$, which is similar to the complex case. Based on this, the norm of q is defined as $|q| = \sqrt{qq^*} = \sqrt{[q]_0^2 + [q]_1^2 + [q]_2^2 + [q]_3^3}$. Let $p, q \in \mathbb{H}$, it can easily verify that $(qp)^* = p^* q^*$, $|qp| = |q||p|$, $\forall p, q \in \mathbb{H}$. Using the conjugate and the norm of q, the inverse of $q \in \mathbb{H}$ is defined by $q^{-1} = \frac{q^*}{|q|^2}$, which indicates that $\mathbb{H}$ is a normed division algebra [23].

In quaternionic notation, let $f : \mathbb{R}^2 \to \mathbb{H}$ is a quaternion function, then $f(\mathbf{t}) = [f(\mathbf{t})]_0 + \mathbf{i}[f(\mathbf{t})]_1 + \mathbf{j}[f(\mathbf{t})]_2 + \mathbf{k}[f(\mathbf{t})]_3$, where $\mathbf{t} = (t_1, t_2) \in \mathbb{R}^2$. The quaternion module $L^2(\mathbb{R}^2, \mathbb{H})$ is defined as $L^2(\mathbb{R}^2, \mathbb{H}) = \{f \mid f : \mathbb{R}^2 \to \mathbb{H}, ||f|| = \left(\int_{\mathbb{R}^2} |f(\mathbf{t})|^2 \mathrm{d}\mathbf{t}\right)^{1/2} < \infty\}$, where $\mathrm{d}\mathbf{t} = \mathrm{d}t_1 \mathrm{d}t_2$.

Consider two quaternion functions $f, g \in L^2(\mathbb{R}^2, \mathbb{H})$, an inner product of f, g as follows $(f, g)_{L^2(\mathbb{R}^2,\mathbb{H})} = \int_{\mathbb{R}^2} f(\mathbf{t}) g^*(\mathbf{t}) \mathrm{d}\mathbf{t}$ with symmetric real scalar part $< f, g >_{L^2(\mathbb{R}^2,\mathbb{H})} = \int_{\mathbb{R}^2} [f(\mathbf{t}) g^*(\mathbf{t})]_0 \mathrm{d}\mathbf{t}$ whose associated norm is $||f||_{L^2(\mathbb{R}^2,\mathbb{H})} = \sqrt{(f, f)_{L^2(\mathbb{R}^2,\mathbb{H})}} = \sqrt{< f, f >_{L^2(\mathbb{R}^2,\mathbb{H})}} = \left(\int_{\mathbb{R}^2} |f(\mathbf{t})|^2 \mathrm{d}\mathbf{t}\right)^{1/2}$. Furthermore, the quaternionic Cauchy-Schwarz inequality is $|(f, g)| \leq ||f||_{L^2(\mathbb{R}^2,\mathbb{H})} ||g||_{L^2(\mathbb{R}^2,\mathbb{H})}$.

2.2 Quaternion linear canonical transform

In this section, we briefly discuss the definition of QLCTs [26] and their important properties.

Definition 2 Let $A_i = (a_i, b_i; c_i, d_i)$ with $a_i d_i - b_i c_i = 1, b_i \neq 0, i = 1, 2$. The two-sided QLCT (TQLCT), left-sided QLCT (LQLCT), and right-sided QLCT (RQLCT) of a signal $f \in L^1(\mathbb{R}^2; \mathbb{H})$ are given respectively

$$L^T_{\mathbf{A}}[f](\mathbf{u}) = \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(t_1, u_1) f(\mathbf{t}) K^{\mathbf{j}}_{A_2}(t_2, u_2) \mathrm{d}\mathbf{t},$$
$$L^L_{\mathbf{A}}[f](\mathbf{u}) = \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(t_1, u_1) K^{\mathbf{j}}_{A_2}(t_2, u_2) f(\mathbf{t}) \mathrm{d}\mathbf{t},$$
$$L^R_{\mathbf{A}}[f](\mathbf{u}) = \int_{\mathbb{R}^2} f(\mathbf{t}) K^{\mathbf{i}}_{A_1}(t_1, u_1) K^{\mathbf{j}}_{A_2}(t_2, u_2) \mathrm{d}\mathbf{t},$$

where $\mathbf{u} = (u_1, u_2), \mathbf{t} = (t_1, t_2), \mathbf{I} = (\mathbf{i}, \mathbf{j}), \mathbf{A} = (A_1, A_2)$ and $K^{\mathbf{I}_i}_{A_i}(t_i, u_i) = \frac{1}{\sqrt{2\pi b_i}} e^{\mathbf{I}_i\left(\frac{a_i}{2b_i}t_i^2 - \frac{1}{b_i}t_i u_i + \frac{d_i}{2b_i}u_i^2\right)}$.

It is sufficient to consider the TQLCT because homologous results for LQLCT or RQLCT can be inferred from their relationships in [23]. For the sake of simplicity, $F^f_{\mathbf{A}}(\mathbf{u})$ is used to represent the TQLCT of f, defined as $F^f_{\mathbf{A}}(\mathbf{u}) := L^T_{\mathbf{A}}[f](\mathbf{u})$.

Definition 3 Suppose $f \in L^2(\mathbb{R}^2; \mathbb{H})$, then the inversion of the TQLCT of f is given by

$$\begin{aligned} f(\mathbf{t}) &= L^{-1}_{\mathbf{A}}[L^T_{\mathbf{A}}[f]](\mathbf{t}) \\ &= \int_{\mathbb{R}^2} K^{-i}_{A_1}(t_1, u_1) L^T_{\mathbf{A}}[f](\mathbf{u}) K^{-j}_{A_2}(t_2, u_2) \mathrm{d}\mathbf{u}. \end{aligned} \tag{2}$$

2.3 Quaternion Wigner–Ville distribution

Let the tensor product of f, g be $(f \otimes g)(\mathbf{x}, \mathbf{y}) = f(\mathbf{x})g(\mathbf{y})$ and add the offset item $\overset{\boldsymbol{\tau}/2}{\otimes}$ on variables, which changed as $(f \overset{\boldsymbol{\tau}/2}{\otimes} g)(\mathbf{x}, \mathbf{y}) = f(\mathbf{x} + \boldsymbol{\tau}/2) g(\mathbf{y} - \boldsymbol{\tau}/2)$, where $\mathbf{x} = (x_1, x_2), \mathbf{y} = (y_1, y_2), \boldsymbol{\tau} = (\tau_1, \tau_2)$. Therefore, the instantaneous autocorrelation function of a function $f(\mathbf{t})$ is defined as

$$(f \overset{\boldsymbol{\tau}/2}{\otimes} f^*)(\mathbf{t}) = f\left(\mathbf{t} + \frac{\boldsymbol{\tau}}{2}\right) f^*\left(t - \frac{\boldsymbol{\tau}}{2}\right). \tag{3}$$

The quaternion Wigner–Ville distributions (QWDs) are defined as follows [22].

Definition 4 The two-sided QWD (TQWD) of a signal $f \in L^1(\mathbb{R}^2; \mathbb{H})$ is given by

$$W^T[f](\mathbf{t}, \mathbf{u}) = \int_{\mathbb{R}^2} e^{-\mathbf{i}u_1\tau_1} (f \overset{\boldsymbol{\tau}/2}{\otimes} f^*)(\mathbf{t}) e^{-\mathbf{j}u_2\tau_2} \mathrm{d}\boldsymbol{\tau},$$

where $\boldsymbol{\tau} = (\tau_1, \tau_2)$ and $\mathrm{d}\boldsymbol{\tau} = \mathrm{d}\tau_1 \mathrm{d}\tau_2$.

3 The PQWD and PQAF

3.1 Definition

In this section, a new parametric Wigner–Ville distribution and a new parametric ambiguity function associated with TQLCT are proposed. They are denoted by PQWD and PQAF, respectively.

The PQWD is generated by deforming the offset tensor product $\overset{\boldsymbol{\tau}/2}{\otimes}$ to the general form $\overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes}$. The instantaneous correlation function is also changed as follows:

$$\left(f \overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes} f^*\right)(\mathbf{t}) = f\left(\boldsymbol{t} + \frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right) f^*\left(\boldsymbol{t} - \frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right). \tag{4}$$

In addition to the scale factor $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2)$, we further consider the parametric embedding of the IAF, which increases flexibility.

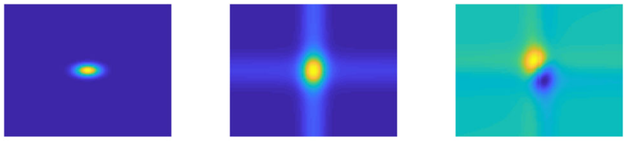
Definition 5 (PQWD) Let $A_i = (a_i, b_i; c_i, d_i)$ with $a_i d_i - b_i c_i = 1, b_i \neq 0, i = 1, 2$, and $B_j = (a_j{}', b_j{}'; c_j{}', d_j{}')$ with $a_i{}'d_i{}' - b_i{}'c_i{}' = 1, b_i{}' \neq 0, j = 1, 2$, represent the linear canonical parameters. The PQWD of a signal $f \in L^2(\mathbb{R}^2; \mathbb{H})$ is given by

$$\begin{aligned} \mathscr{W}^{\epsilon,T}_{\mathbf{A},\mathbf{B}}[f](\mathbf{t}, \mathbf{u}) = \int_{\mathbb{R}^2} & K^{\mathbf{i}}_{B_1}(\tau_1, u_1) \left(F^f_{\mathbf{A}} \overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes} f^*\right)(\mathbf{t}) \\ & \times K^{\mathbf{j}}_{B_2}(\tau_2, u_2) \mathrm{d}\boldsymbol{\tau}, \end{aligned} \tag{5}$$

where $F^f_{\mathbf{A}}(\mathbf{u})$ is the TQLCT for f with respect to $\boldsymbol{\tau}, \mathbf{u} = (u_1, u_2), \boldsymbol{\tau} = (\tau_1, \tau_2), \mathbf{t} = (t_1, t_2), \boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2), \mathbf{A} = (A_1, A_2), \mathbf{B} = (B_1, B_2), \mathbf{I} = (\mathbf{i}, \mathbf{j}), \boldsymbol{\tau}/\boldsymbol{\epsilon} = (\tau_1/\epsilon_1, \tau_2/\epsilon_2)$, and

$$K^{\mathbf{I}_i}_{A_i}(t_i, u_i) = \frac{e^{\mathbf{I}_i\left(\frac{a_i}{2b_i}t_i^2 - \frac{1}{b_i}t_i u_i + \frac{d_i}{2b_i}u_i^2\right)}}{\sqrt{2\pi b_i}}, \tag{6}$$

$$K^{\mathbf{I}_j}_{B_j}(t_j, u_j) = \frac{e^{\mathbf{I}_j\left(\frac{a'_j}{2b'_j}t_j^2 - \frac{1}{b'_j}t_j u_j + \frac{d'_j}{2b'_j}u_j^2\right)}}{\sqrt{2\pi b'_j}}. \tag{7}$$



(a) (b) (c)

Fig. 1 **a** The 2-D Gaussian signal $f(t_1,t_2)$ in (11). **b** Real-part of $\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](-0.0202,-0.0202,u_1,u_2)$. **c** Imaginary-part of $\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](-0.0202,-0.0202,u_1,u_2)$

Definition 6 (PQAF) The PQAF of a signal $f\in L^2(\mathbb{R}^2;\mathbb{H})$ is given by

$$\mathscr{A}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](\boldsymbol{\tau},\mathbf{u})=\int_{\mathbb{R}^2} K^{\mathbf{i}}_{B_1}(t_1,u_1)\left(F^f_{\mathbf{A}}\overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes} f^*\right)(\mathbf{t}) \times K^{\mathbf{j}}_{B_2}(t_2,u_2)\mathrm{d}\mathbf{t}, \tag{8}$$

where $F^f_{\mathbf{A}}(\mathbf{u})$ is the TQLCT for f with respect to $\mathbf{t}$, $K^{\mathbf{I}_i}_{A_i}(t_i,u_i)$, $K^{\mathbf{I}_j}_{B_j}(t_j,u_j)$ are identical to (6) and (7).

The following theorem establishes the relationship between PQWD and PQAF, demonstrating that they can be transformed into one another. Therefore, without loss of generality, we will focus solely on PQWD for the remainder of this paper.

Theorem 1 *For $f\in L^2(\mathbb{R}^2,\mathbb{H})$, we have*

$$\mathscr{A}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](\boldsymbol{\tau},\mathbf{u})=\frac{1}{\epsilon_1\epsilon_2}\mathscr{W}^{\boldsymbol{\epsilon},T}_{-\mathbf{A},\tilde{\mathbf{B}}}[f]\left(-\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}},\frac{\mathbf{u}}{\boldsymbol{\epsilon}}\right), \tag{9}$$

$$\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](\boldsymbol{\tau},\mathbf{u})=\epsilon_1\epsilon_2\mathscr{A}^{\boldsymbol{\epsilon},T}_{-\mathbf{A},\hat{\mathbf{B}}}[f](\boldsymbol{\tau},\mathbf{u}\boldsymbol{\epsilon}), \tag{10}$$

where $\tilde{\mathbf{B}}=(\tilde{B}_1,\tilde{B}_2)$, $\tilde{B}_i=(-a_i'/{\epsilon_i}^2,-b_i';-c_i',-d_i'\epsilon_i^2)$, and $\hat{\mathbf{B}}=(\hat{B}_1,\hat{B}_2)$, $\hat{B}_i=(-{\epsilon_i}^2a_i',-b_i';-c_i',-d_i'/\epsilon_i^2)$, $i=1,2$.

Additionally, we demonstrate the application of PQWD on a two-dimensional Gaussian signal in Example 1.

Example 1 Consider a 2-D Gaussian signal of the form

$$f(t_1,t_2)=e^{-\pi\left(\frac{t_1^2}{0.5^2}+\frac{t_2^2}{0.3^2}\right)}, \tag{11}$$

where $t_1,t_2\in[-2,2]$, and the signal size is 100×100. Figure 1 displays the PQWD $\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](-0.0202,-0.0202,u_1,u_2)$ of a 2-D Gaussian signal with $A_1=A_2=(0,1;-1,0)$, $B_1=B_2=(2,0.9999;-1/0.9999,0)$, $\epsilon_1=\epsilon_2=3$.

3.2 Properties

Several important properties of the PQWD are discussed in this subsection, and the detailed proofs are presented. These properties play critical roles in signal representation.

Theorem 2 (Anti-derivative) *Let $f\in L^2(\mathbb{R}^2;\mathbb{H})$ and $f(\mathbf{0})\neq\mathbf{0}$, $F^f_{\mathbf{A}}(\mathbf{0})\neq\mathbf{0}$, then the f and the LCT of f can be expressed by the PQWD, respectively. That is*

$$f(\mathbf{t})=\frac{1}{[F^f_{\mathbf{A}}(0)]^*}\int_{\mathbb{R}^2}\left[K^{\mathbf{i}}_{B_1^{-1}}\left(-\frac{t_1\epsilon_1}{2},u_1\right)\times\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f]\left(\frac{\mathbf{t}}{2},\mathbf{u}\right)K^{\mathbf{j}}_{B_2^{-1}}\left(-\frac{t_2\epsilon_2}{2},u_2\right)\right]^*\mathrm{d}\mathbf{u},$$

and

$$F^f_{\mathbf{A}}(\mathbf{t})=\frac{1}{f^*(0)}\int_{\mathbb{R}^2}K^{\mathbf{i}}_{B_1^{-1}}\left(\frac{t_1\epsilon_1}{2},u_1\right)\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f]\left(\frac{\mathbf{t}}{2},\mathbf{u}\right)K^{\mathbf{j}}_{B_2^{-1}}\left(\frac{t_2\epsilon_2}{2},u_2\right)\mathrm{d}\mathbf{u}.$$

Theorem 3 (Time shift) *Let $f\in L^2(\mathbb{R}^2;\mathbb{H})$, consider the operator of time shifting $T_{\boldsymbol{\xi}}f(\mathbf{t})=f(\mathbf{t}-\boldsymbol{\xi})$, we have*

$$\left[\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[T_{\boldsymbol{\xi}}f](\mathbf{t},\mathbf{u})\right]_0=e^{\mathbf{i}\left(\frac{d_1u_1}{\epsilon_1}+t_1\right)c_1\xi_1} e^{-\mathbf{i}\left(\frac{b_1'd_1'c_1}{2\epsilon_1^2}+\frac{a_1c_1}{2}\right)c_1\xi_1^2}\left[\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f]\left(\mathbf{t}-\boldsymbol{\xi},\mathbf{u}-\frac{\mathbf{b}'\mathbf{c}}{\boldsymbol{\epsilon}}\boldsymbol{\xi}\right)\right]_0 e^{\mathbf{j}\left(\frac{d_2u_2}{\epsilon_2}+t_2\right)c_2\xi_2}e^{-\mathbf{j}\left(\frac{b_2'd_2'c_2}{2\epsilon_2^2}+\frac{a_2c_2}{2}\right)c_2\xi_2^2},$$

where $\mathbf{A}=(A_1,A_2)$, $\mathbf{B}=(B_1,B_2)$, $\mathbf{I}=(\mathbf{i},\mathbf{j})$, $A_i=(1,b_i;c_i,d_i)$, $\mathbf{u}=(u_1,u_2)$, $\boldsymbol{\xi}=(\xi_1,\xi_2)$, $\mathbf{a}=(a_1,a_2)$, $\mathbf{b}'=(b_1',b_2')$.

Theorem 4 (Frequency shift) *Let $f\in L^2(\mathbb{R}^2;\mathbb{H})$, applying the frequency shift operator $S_{\boldsymbol{\xi}}f(\mathbf{t})=e^{\mathbf{i}\xi_1t_1}f(\mathbf{t})e^{\mathbf{j}\xi_2t_2}$, then*

$$[\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[S_{\boldsymbol{\xi}}f](\mathbf{t},\mathbf{u})]_0=e^{-\mathbf{i}\left(\frac{b_1d_1}{2}+\frac{b_1'^2(d_1+1)^2}{\epsilon_1^2}\right)\xi_1^2} e^{\mathbf{i}\left((d_1-1)t_1+\frac{d_1'(d_1+1)u_1}{\epsilon_1}\right)\xi_1}\left[\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f]\left(\mathbf{t},\mathbf{u}-\frac{\mathbf{d}+1}{\boldsymbol{\epsilon}}\mathbf{b}'\boldsymbol{\xi}\right)\right]_0 e^{-\mathbf{j}\left(\frac{b_2d_2}{2}+\frac{b_2'^2(d_2+1)^2}{\epsilon_2^2}\right)\xi_2^2}e^{\mathbf{j}\left((d_2-1)t_2+\frac{d_2'(d_2+1)u_2}{\epsilon_2}\right)\xi_2},$$

where $\mathbf{A}=(A_1,A_2)$, $\mathbf{B}=(B_1,B_2)$, $\mathbf{I}=(\mathbf{i},\mathbf{j})$, $A_i=(a_i,0;c_i,d_i)$, $\mathbf{u}=(u_1,u_2)$, $B_j=(a_j',b_j';c_j',d_j')$, $\boldsymbol{\xi}=(\xi_1,\xi_2)$, $\mathbf{b}'=(b_1',b_2')$, $\mathbf{d}=(d_1,d_2)$.

Theorem 5 (Boundedness) *Let* $f \in L^2(\mathbb{R}^2;\mathbb{H})$*, then the* $\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})$ *is bounded on* $L^4(\mathbb{R}^2;\mathbb{H})$*, that is*

$$\left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right| \leq \sqrt{\frac{\epsilon_1^2\epsilon_2^2}{8\pi^3|b_1'b_2'|\sqrt{|b_1b_2|}}} \times ||f(t)||^2_{L^2(\mathbb{R}^2;\mathbb{H})}. \tag{12}$$

Theorem 6 (Parseval theorem) *Let* $f \in L^2(\mathbb{R}^2;\mathbb{H})$*, then*

$$||\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}||_{L^2(\mathbb{R}^2;\mathbb{H})} = ||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})}. \tag{13}$$

4 Uncertainty principles for the PQWD

The uncertainty principle is a cornerstone of mathematics, signal processing, and physics. It reveals that parameter adjustments can alter the lower bounds of uncertainty, enabling optimized time and frequency resolutions for various scenarios. While extensive studies have focused on the uncertainty principles of transforms like the LCT, WD, and their variants [14, 15, 25], much remains to be explored in this field.

4.1 Heisenberg's uncertainty principle

In this section, we provide a brief and straightforward proof of the Heisenberg's uncertainty principle of the PQWD. The Heisenberg's uncertainty principle for the TQLCT in [30] is

$$\int_{\mathbb{R}^2} t_k^2|f(\mathbf{t})|^2\mathrm{d}\mathbf{t}\int_{\mathbb{R}^2} u_k^2|L_{\mathbf{A}}^T[f](\mathbf{u})|^2\mathrm{d}\mathbf{u} \geq \frac{b_k^2}{4}\left(\int_{\mathbb{R}^2}|f(\mathbf{t})|^2\mathrm{d}\mathbf{t}\right)^2, \tag{14}$$

where $f \in L^2(\mathbb{R}^2;\mathbb{H})$, $k = 1, 2$. Now we arrive at the following important result.

Theorem 7 (Heisenberg's uncertainty principle for the PQWD) *Let* $f \in L^2(\mathbb{R}^2;\mathbb{H})$*, and* $\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u}) \in L^2(\mathbb{R}^2;\mathbb{H})$ *be the PQWD of* f*. Then the following inequality is held*

$$\int_{\mathbb{R}^2} t_k^2\,|f(\mathbf{t})|^2\,\mathrm{d}\mathbf{t}\int_{\mathbb{R}^4} u_k^2\left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right|^2\mathrm{d}\mathbf{u}\,\mathrm{d}\mathbf{t} \geq \frac{b_k'^2}{4}||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})}.$$

Proof Substituting the inverse transform for the TQLCT (2) into the left side of (14), we obtain

$$\int_{\mathbb{R}^2} t_k^2|L_{\mathbf{A}}^{-1}[L_{\mathbf{A}}^T[f]](\mathbf{t})|^2\mathrm{d}\mathbf{t}\int_{\mathbb{R}^2} u_k^2|L_{\mathbf{A}}^T[f](\mathbf{u})|^2\mathrm{d}\mathbf{u} \geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2}|f(\mathbf{t})|^2\mathrm{d}\mathbf{t}\right)^2,$$

where $k = 1, 2$. Moreover, using the Paseval formula for the TQLCT,

$$\int_{\mathbb{R}^2} t_k^2|L_{\mathbf{A}}^{-1}[L_{\mathbf{A}}^T[f]](\mathbf{t})|^2\mathrm{d}\mathbf{t}\int_{\mathbb{R}^2} u_k^2|L_{\mathbf{A}}^T[f](\mathbf{u})|^2\mathrm{d}\mathbf{u} \geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2}|L_{\mathbf{A}}^T[f](\mathbf{u})|^2\mathrm{d}\mathbf{u}\right)^2.$$

Since $\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u}) \in L^2(\mathbb{R}^2;\mathbb{H})$, we can replace the TQLCT of f by the PQWD of f, we have

$$\int_{\mathbb{R}^2} \tau_k^2|L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2\mathrm{d}\boldsymbol{\tau}\int_{\mathbb{R}^2} u_k^2|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u} \geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u}\right)^2.$$

Integrating both sides with respect to d$\mathbf{t}$,

$$\int_{\mathbb{R}^2}\left(\int_{\mathbb{R}^2}\tau_k^2|L_{\mathbf{A}}^{-1}\left[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right](\boldsymbol{\tau})|^2\mathrm{d}\boldsymbol{\tau}\right)\left(\int_{\mathbb{R}^2}u_k^2|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u}\right)\mathrm{d}\mathbf{t} \geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u}\right)^2.$$

Further, applying the Cauchy-Schwarz inequality, we obtain

$$\left(\int_{\mathbb{R}^4}\tau_k^2|L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2\mathrm{d}\boldsymbol{\tau}\,\mathrm{d}\mathbf{t}\right) \times\left(\int_{\mathbb{R}^4}u_k^2|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u}\,\mathrm{d}\mathbf{t}\right) \geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2\mathrm{d}\mathbf{u}\right)^2. \tag{15}$$

Based on the Paseval formula for the TQLCT, we have

$$\begin{aligned}&\int_{\mathbb{R}^4} \tau_k^2 |L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2 \mathrm{d}\boldsymbol{\tau}\,\mathrm{d}\mathbf{t}\\ &= \int_{\mathbb{R}^4} \tau_k^2 \left|F_{\mathbf{B}}^f\left(\mathbf{t}+\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \left|f\left(\mathbf{t}-\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}\,\mathrm{d}\mathbf{t}\\ &= \int_{\mathbb{R}^4} \tau_k^2 \left|F_{\mathbf{B}}^f(\mathbf{t})\right|^2 \left|f\left(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}\,\mathrm{d}\mathbf{t}\\ &= ||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})} \int_{\mathbb{R}^4} \tau_k^2 \left|f\left(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}. \end{aligned} \tag{16}$$

And then taking (16) into (15), we get

$$\begin{aligned}&\left(||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})} \int_{\mathbb{R}^4} \tau_k^2 \left|f\left(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}\right)\\ &\left(\int_{\mathbb{R}^4} u_k^2 \left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right|^2 \mathrm{d}\mathbf{u}\,\mathrm{d}\mathbf{t}\right)\\ &\geq \frac{b_k'^2}{4}\left(\int_{\mathbb{R}^2} |\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 \mathrm{d}\mathbf{u}\right)^2. \end{aligned} \tag{17}$$

By organizing the above equation, we can obtain the final result. □

4.2 Logarithmic uncertainty principle

The Heisenberg's uncertainty principlehas been generalized in various ways, including the logarithmic uncertainty principle. The logarithmic uncertainty principles of the LCT and the quaternion LCT have been previously studied in citations [28, 31, 32]. In this subsection, we aim to derive the logarithmic uncertainty principle of the PQWD.

Bahri introduced the logarithmic uncertainty principle for the TQLCT [32]. Let $f \in L^2(\mathbb{R}^2;\mathbb{H})$, and $L_{\mathbf{A}}^T[f](\mathbf{u}) \in L^2(\mathbb{R}^2;\mathbb{H})$ be the TQLCT of f. Then

$$\begin{aligned}&\int_{\mathbb{R}^2} \ln\sqrt{t_1^2+t_2^2}|f(\mathbf{t})|^2 \mathrm{d}\mathbf{t} + \int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|L_{\mathbf{A}}^T[f](\mathbf{u})|^2 \mathrm{d}\mathbf{u}\\ &\geq \left(D+\ln\sqrt{b_1^2+b_2^2}\right)\int_{\mathbb{R}^2} |f(\mathbf{t})|^2 \mathrm{d}\mathbf{t}, \end{aligned} \tag{18}$$

where $D = \psi\left(\frac{1}{2}\right)$, $\psi(t) = \frac{d}{dt}[\Gamma(\mathrm{t})]$, and $\Gamma(t)$ is the gamma function.

To derive the logarithmic uncertainty principle of PQWD, we will use the following lemma.

Lemma 1 *Let $f \in L^2(\mathbb{R}^2;\mathbb{H})$, then*

$$\begin{aligned}&\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}|L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2 \mathrm{d}\mathbf{t}\,\mathrm{d}\boldsymbol{\tau}\\ &= ||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})} \int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}\left|f\left(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}. \end{aligned} \tag{19}$$

Then we focus on the following logarithmic uncertainty principle of the PQWD.

Theorem 8 (Logarithmic uncertainty principle for the PQWD) *Let $f \in L^2(\mathbb{R}^2;\mathbb{H})$, and $\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u}) \in L^2(\mathbb{R}^2;\mathbb{H})$ be the PQWD of f. Then we have the following inequality*

$$\begin{aligned}&||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})} \int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}\left|f\left(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)\right|^2 \mathrm{d}\boldsymbol{\tau}\\ &\quad+\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 \mathrm{d}\mathbf{u}\,\mathrm{d}\mathbf{t}\\ &\geq \left(D+\ln\sqrt{b_1'^2+b_2'^2}\right)||f||^4_{L^2(\mathbb{R}^2;\mathbb{H})}. \end{aligned} \tag{20}$$

Proof By applying the inverse transform (2) and the Paseval formula for the TQLCT to (18), we have

$$\begin{aligned}&\int_{\mathbb{R}^2} \ln\sqrt{t_1^2+t_2^2}|L_{\mathbf{A}}^{-1}[L_{\mathbf{A}}^T[f]](\mathbf{t})|^2 \mathrm{d}\mathbf{t}\\ &\quad+\int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|L_{\mathbf{A}}^T[f](\mathbf{u})|^2 \mathrm{d}\mathbf{u}\\ &\geq \left(D+\ln\sqrt{b_1^2+b_2^2}\right)\int_{\mathbb{R}^2} |L_{\mathbf{A}}^T[f](\mathbf{u})|^2 \mathrm{d}\mathbf{u}. \end{aligned}$$

Since $\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u}) \in L^2(\mathbb{R}^2;\mathbb{H})$, we can replace the TQLCT of f by the PQWD of f, we get

$$\begin{aligned}&\int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}|L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2 \mathrm{d}\boldsymbol{\tau}\\ &\quad+\int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 \mathrm{d}\mathbf{u}\\ &\geq \left(D+\ln\sqrt{b_1'^2+b_2'^2}\right)\int_{\mathbb{R}^2} |\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 \mathrm{d}\mathbf{u}. \end{aligned} \tag{21}$$

Integrating both sides concerning $\mathbf{dt}$, and applying the Cauchy-Schwarz inequality to the left side of (21), we obtain

$$\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}|L_{\mathbf{A}}^{-1}[\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})](\boldsymbol{\tau})|^2 \mathrm{d}\boldsymbol{\tau}\,\mathrm{d}\mathbf{t}$$

$$+\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 d\mathbf{u}\, d\mathbf{t}$$
$$\geq \left(D+\ln\sqrt{b_1'^2+b_2'^2}\right)\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} |\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 d\mathbf{u}\, d\mathbf{t}.$$

Then substituting the Lemma 1 and Theorem 6, we have

$$||f||^2_{L^2(\mathbb{R}^2;\mathbb{H})}\int_{\mathbb{R}^2} \ln\sqrt{\tau_1^2+\tau_2^2}|f(\mathbf{t}-\frac{2\boldsymbol{\tau}}{\boldsymbol{\epsilon}})|^2 d\boldsymbol{\tau}$$
$$+\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \ln\sqrt{u_1^2+u_2^2}|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^2 d\mathbf{u}\, d\mathbf{t}$$
$$\geq \left(D+\ln\sqrt{b_1'^2+b_2'^2}\right)||f||^4_{L^2(\mathbb{R}^2;\mathbb{H})}.$$

Which completes the proof. □

4.3 Lieb's uncertainty principle

This subsection presents Lieb's uncertainty principle for the PQWD. Let us first review the concept of σ-concentrated of a quaternion signal function on any measurable set $M \subseteq \mathbb{R}^2$.

Definition 7 A quaternion signal $f \in L^2(\mathbb{R}^2;\mathbb{H})$ is said to be σ-concentrated [28] on a measurable set $M \subseteq \mathbb{R}^2$, if

$$\left(\int_{\mathbb{R}^2\setminus M} |f(\mathbf{t})|^2 d\mathbf{t}\right)^{\frac{1}{2}} \leq \sigma ||f||_{L^2(\mathbb{R}^2;\mathbb{H})}. \tag{22}$$

Moreover, the Hausdorff-Young inequality for the TQLCT is necessary to derive the Lieb's uncertainty principle.

Lemma 2 (Hausdorff-Young inequality) *[28] Let $\frac{1}{p}+\frac{1}{q}=1$ for $1 \leq p \leq 2$. Then for every $L^T_{\mathbf{B}}[f] \in L^p(\mathbb{R}^2;\mathbb{H})$, there holds*

$$||L^T_{\mathbf{A}}[f](\mathbf{u})||_q \leq (2\pi)^{\frac{1}{q}-\frac{1}{p}}|b_1''b_2''|^{\frac{1}{2}-\frac{1}{p}}||L^T_{\mathbf{B}}[f](\mathbf{u})||_p,$$

where $\mathbf{A}=(\mathbf{A}_1,\mathbf{A}_2)$, $\mathbf{A}_i=(a_i,b_i;c_i;d_i)$, $\mathbf{B}=(\mathbf{B}_1,\mathbf{B}_2)$, $\mathbf{b}_j=(a_j',b_j';c_j';d_j')$, $b_1''=a_1'b_1-a_1b_1'$, $b_2''=a_2'b_2-a_2b_2'$.

Theorem 9 (Lieb's uncertainty principle for the PQWD) *Let $f \in L^2(\mathbb{R}^2;\mathbb{H})$, and $\frac{1}{p}+\frac{1}{q}=1$ for $2 \leq p \leq 2$. Then*

$$\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right|^q d\mathbf{u}\, d\mathbf{t} \leq (2\pi)^{1-\frac{q}{p}}$$
$$|b_1'''b_2'''|^{\frac{q}{2}-\frac{q}{p}}\left(\frac{4}{q}\right)\left(\frac{1}{p}\right)^{\frac{q}{p}}||f||^{2q}_{L^2(\mathbb{R}^2;\mathbb{H})}, \tag{23}$$

where $\mathbf{A}=(\mathbf{A}_1,\mathbf{A}_2)$, $\mathbf{B}=(\mathbf{B}_1,\mathbf{B}_2)$, $\mathbf{C}=(\mathbf{C}_1,\mathbf{C}_2)$, $\mathbf{A}_i=(a_i,b_i;c_i;d_i)$, $\mathbf{B}_j=(a_j',b_j';c_j';d_j')$, $\mathbf{C}_k=(a_k'',b_k'';c_k'';d_k'')$, $b_1'''=a_1''b_1'-a_1'b_1''$, $b_2'''=a_2''b_2'-a_2'b_2''$, and $C_1=C_2=(1,0;0,1)$.

Proof By applying Lemma 2, we have

$$||L^T_{\mathbf{B}}[F^f_{\mathbf{A}} \overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes} f^*](\mathbf{u})||_q \leq (2\pi)^{\frac{1}{q}-\frac{1}{p}}|b_1'''b_2'''|^{\frac{1}{2}-\frac{1}{p}}||$$
$$L^T_{\mathbf{C}}[F^f_{\mathbf{A}} \overset{\boldsymbol{\tau}/\boldsymbol{\epsilon}}{\otimes} f^*](\mathbf{u})||_p.$$

Using the Parseval formula for the TQLCT, the above inequality equals to

$$||\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})||_q \leq (2\pi)^{\frac{1}{q}-\frac{1}{p}}|b_1'''b_2'''|^{\frac{1}{2}-\frac{1}{p}}$$
$$||F^f_{\mathbf{A}}\left(\mathbf{t}+\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)f^*\left(\mathbf{t}-\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)||_p.$$

Integrating both sides of the above equation for $d\mathbf{t}$ and using the Holder inequality, we get

$$\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right|^q d\mathbf{u}\, d\mathbf{t}\right)^{\frac{1}{q}}$$
$$\leq (2\pi)^{\frac{1}{q}-\frac{1}{p}}|b_1'''b_2'''|^{\frac{1}{2}-\frac{1}{p}}$$
$$\times\left(\int_{\mathbb{R}^2}\left(\int_{\mathbb{R}^2}|F^f_{\mathbf{A}}\left(\mathbf{t}+\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)f^*\left(\mathbf{t}-\frac{\boldsymbol{\tau}}{\boldsymbol{\epsilon}}\right)|^p d\mathbf{t}\right)^{\frac{q}{p}} d\boldsymbol{\tau}\right)^q.$$

And applying the Young's inequality in [33], we obtain

$$\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} |\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})|^q d\mathbf{u}\, d\mathbf{t}\right)^{\frac{1}{q}}$$
$$\leq (2\pi)^{\frac{1}{q}-\frac{1}{p}}|b_1'''b_2'''|^{\frac{1}{2}-\frac{1}{p}}\left(\frac{4}{q}\right)^{\frac{1}{q}}\left(\frac{1}{p}\right)^{\frac{1}{p}}||F^f_{\mathbf{A}}||_{L^2(\mathbb{R}^2;\mathbb{H})}$$
$$||f||_{L^2(\mathbb{R}^2;\mathbb{H})}.$$

Then taking the q root on both sides and applying the Parseval formula for the TQLCT,

$$\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \left|\mathscr{W}_{\mathbf{A},\mathbf{B}}^{\boldsymbol{\epsilon},T}[f](\mathbf{t},\mathbf{u})\right|^q d\mathbf{u}\, d\mathbf{t}$$
$$\leq (2\pi)^{1-\frac{q}{p}}|b_1'''b_2'''|^{\frac{q}{2}-\frac{q}{p}}\left(\frac{4}{q}\right)\left(\frac{1}{p}\right)^{\frac{q}{p}}||f||^{2q}_{L^2(\mathbb{R}^2;\mathbb{H})},$$

which completes the proof. □

Then we can obtain a lower bound for the essential support of the PQWD.

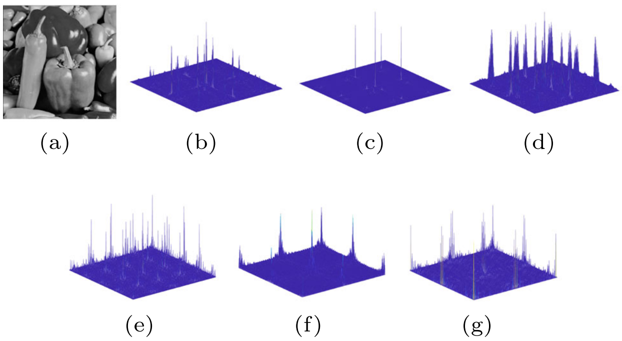


Fig. 2 **a** Peppers image. **b** $t_1 = t_2 = 64$. **c** $t_1 = t_2 = 256$. **d** $t_1 = t_2 = 480$. **e** $B_1 = B_2 = (0.5, 2; -1, -2)$. **f** $B_1 = B_2 = (1, 1.5; -1, -0.5)$. **g** $B_1 = B_2 = (2, 4.5; -1, -1.75)$

Corollary 1 *Let* $f \in L^2(\mathbb{R}^2; \mathbb{H})$ *with* $||f||_{L^2(\mathbb{R}^2;\mathbb{H})} = 1$. *If* $\mathscr{W}^{\boldsymbol{\epsilon},T}_{\mathbf{A},\mathbf{B}}[f](\mathbf{t}, \mathbf{u})$ *is* σ*-concentrated on a measurable set* $M \subseteq \mathbb{R}^2 \times \mathbb{R}^2$ *for* $\sigma \geq 0$, *and* $\frac{1}{p} + \frac{1}{q} = 1$ *for* $2 \leq p \leq 2$. *Then*

$$|M| \geq (2\pi)^2 |b_1''' b_2'''| \left(\left(\frac{4}{q}\right)^{\frac{1}{q}} \left(\frac{1}{p}\right)^{\frac{1}{p}} \right)^{\frac{2q}{2-q}} \left(1 - \sigma^2\right)^{\frac{q}{q-2}}.$$

5 Potential application

In this section, we explore potential applications of our approach. The PQWD is applied to the Peppers image (Fig. 2a) using various (t_1, t_2) coordinates, where $t_1, t_2 = 1, \ldots, 256$, with parameters $A_1 = A_2 = (0, 2; -2, 0)$, $B_1 = B_2 = (3, 0.9; -1/0.9, 0)$, and $\epsilon_1 = \epsilon_2 = 2$. The results (amplitude), illustrated in Fig. 2b–d, demonstrate PQWD's capability to effectively capture distinct image features. By varying the coordinates, a diverse range of local features from different image regions can be extracted. By adjusting the PQWD parameters, different feature sets of the same image can be obtained, as shown in Fig. 2e–f.

Building on PQWD's feature extraction capabilities, we integrate it into a neural network for image classification tasks to evaluate its effectiveness. PQWD is employed as a feature extraction method, transforming input signals into the time-frequency domain to enable the network to learn discriminative representations. The workflow for classification using PQWD is shown in Algorithm 1. The MNIST dataset comprises 70,000 grayscale images of handwritten digits (0–9), each sized 28×28 pixels. For this experiment, we focus on a binary classification task involving digits 0 and 1. A total of 6900 samples per digit are selected to ensure balanced classes.

We evaluate the classification performance of the model under different parameter settings. To simplify the analysis, we set $A_1 = A_2 = (0, 2; -2, 0)$ and $\epsilon_1 = \epsilon_2 = 2$. By adjusting $a_1' = a_2'$ and $b_1' = b_2'$, different parameters B_1 and B_2 are obtained, and the results are presented in Fig. 3a. When $B_1 = B_2 = (1.5, 5, -1, -2.67)$, the model achieves the highest classification accuracy of 97.71%. In contrast, for the simplified model, which corresponds to the traditional QWD, the classification accuracy is 93.24%, demonstrating the flexibility and superiority introduced by parameter embedding. Additionally, we investigate the impact of varying t_1 and t_2 on the model's performance under the fixed parameters $B_1 = B_2 = (1.5, 5, -1, -2.67)$. The results are shown in Fig. 3b. It is observed that the closer t_1 and t_2 are to the central region, the stronger the ability of the PQWD model to capture features. Specifically, when $t_1 = 16$ and $t_2 = 10$, the model exhibits optimal performance. The above results highlight the superiority of the PQWD.

Algorithm 1 Classification Using PQWD

1: **Input:** Image data $\mathbf{X}$, label $\mathbf{y}$.
2: **Output:** Predicted labels $\hat{\mathbf{y}}$.
3: **for** each input signal $\mathbf{f} \in \mathbf{X}$ **do**
4: Compute the PQWD of $\mathbf{f}$ using (5).
5: Flatten the PQWD output into a feature vector.
6: **end for**
7: Train the neural network with a single hidden layer of size 10 on the extracted features and labels .
8: Predict using the trained neural network.

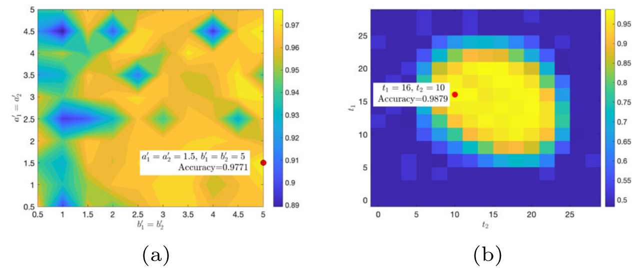


Fig. 3 Classification accuracy. **a** Under different parameters a_1', b_1', a_2', b_2'. **b** Under different t_1,t_2

6 Conclusion

In this paper, we introduce the parametric quaternion Wigner–Ville distribution (PQWD) and explore its fundamental properties. A key contribution of this paper is the presentation of uncertainty principles associated with the PQWD. These findings provide valuable insights that may have significant implications for practical applications. Additionally, we discuss the potential of PQWD in feature extraction, which suggests promising avenues for further research and application in signal processing.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Zhang, Z.C.: Unified Wigner-Ville distribution and ambiguity function in the linear canonical transform domain. Signal Process. **114**, 45–60 (2015)
2. Bhat, M.Y., Dar, A.H.: Convolution and correlation theorems for Wigner-Ville distribution associated with the quaternion offset linear canonical transform. SIViP **16**(5), 1235–1242 (2022)
3. Zhang, Z.C.: New Wigner distribution and ambiguity function based on the generalized translation in the linear canonical transform domain. Signal Process. **118**, 51–61 (2016)
4. Zhang, Z.C.: Novel Wigner distribution and ambiguity function associated with the linear canonical transform. Optik **127**(12), 4995–5012 (2016)
5. Zhang, Z.C., Jiang, X., Qiang, S.Z., Sun, A., Liang, Z.Y., Shi, X.Y., Wu, A.Y.: Scaled Wigner distribution using fractional instantaneous autocorrelation. Optik **237**, 166691 (2021)
6. Chen, J.Y., Li, B.Z.: The short-time Wigner-Ville distribution. Signal Process. **219**, 109402 (2024)
7. Chen, J.Y., Li, B.Z.: Fast numerical calculation of the offset linear canonical transform. JOSA A **40**(3), 427–442 (2023)
8. Lu, L., Ren, W.X., Wang, S.D.: Fractional Fourier transform: Time-frequency representation and structural instantaneous frequency identification. Mech. Syst. Signal Process. **178**, 109305 (2022)
9. Chen, J.Y., Li, B.Z.: The short-time Wigner-Ville distribution associated with linear canonical transform. In: Fifteenth international conference on signal processing systems (ICSPS 2023), vol. 13091, pp. 28–35 (2024). SPIE
10. Bai, R.F., Li, B.Z., Cheng, Q.Y.: Wigner-Ville distribution associated with the linear canonical transform. J. Appl. Math. **2012**(1), 740161 (2012)
11. Tian Wen, C., Bing Zhao, L., Tian Zhou, X.: The ambiguity function associated with the linear canonical transform. EURASIP J. Adv. Signal Process. **2012**(1), 1–14 (2012)
12. Zhang, Z., Luo, M.: New integral transforms for generalizing the Wigner distribution and ambiguity function. IEEE Signal Process. Lett. **22**(4), 460–464 (2014)
13. Guo, W., Chen, J., Fan, D., Zhao, G.: Characterization of boundedness on weighted modulation spaces of τ-Wigner distributions. Int. Math. Res. Not. **2022**(21), 16844–16901 (2022)
14. Zhang, Z., Shi, X.: Kernel function-τ-Wigner distribution associated with the linear canonical transform. IEEE Signal Process. Lett. **29**, 1764–1768 (2022)
15. Zhang, Z., He, Y., Zhang, J., Zhou, C.: The optimal k-Wigner distribution. Signal Process. **199**, 108608 (2022)
16. Hahn, S.L.: Complex signals with single-orthant spectra as boundary distributions of multidimensional analytic functions. Bull. Pol. Ac Tech. **2**(2), 155–161 (2003)
17. Kou, K.I., Liu, M.S., Morais, J.P., Zou, C.: Envelope detection using generalized analytic signal in 2D QLCT domains. Multidimension. Syst. Signal Process. **28**, 1343–1366 (2017)
18. Gao, W.B., Li, B.Z.: Quaternion windowed linear canonical transform of two-dimensional signals. Adv. Appl. Clifford Algebras **30**(1), 1–18 (2020)
19. Bahri, M., Hitzer, E.S., Ashino, R., Vaillancourt, R.: Windowed Fourier transform of two-dimensional quaternionic signals. Appl. Math. Comput. **216**(8), 2366–2379 (2010)
20. Chen, J.Y., Li, B.Z.: Wigner distribution associated with linear canonical transform of generalized 2-d analytic signals. Digital Signal Process. **149**, 104481 (2024)
21. Bahri, M.: Quaternion linear canonical transform application. Glob. J. Pure Appl. Math. **11**(1), 19–24 (2015)
22. Bahri, M.: On two-dimensional quaternion Wigner-Ville distribution. J. Appl. Math. **2014**(1), 139471 (2014)
23. Fan, X.L., Kou, K.I., Liu, M.S.: Quaternion Wigner-Ville distribution associated with the linear canonical transforms. Signal Process. **130**, 129–141 (2017)
24. Busch, P., Heinonen, T., Lahti, P.: Heisenberg's uncertainty principle. Phys. Rep. **452**(6), 155–176 (2007)
25. Zhao, J., Tao, R., Li, Y.L., Wang, Y.: Uncertainty principles for linear canonical transform. IEEE Trans. Signal Process. **57**(7), 2856–2858 (2009)
26. Kou, K.I., Ou, J.Y., Morais, J.: On uncertainty principle for quaternionic linear canonical transform. Abstract Appl. Anal. **2013**, 725952 (2013)
27. Zhang, Y.N., Li, B.Z.: Novel uncertainty principles for two-sided quaternion linear canonical transform. Adv. Appl. Clifford Algebras **28**(1), 1–14 (2018)
28. Zhu, X., Zheng, S.: Uncertainty principles for the two-sided quaternion linear canonical transform. Circuits Syst. Signal Process. **39**(9), 4436–4458 (2020)
29. Urynbassarova, D., El Haoui, Y., Zhang, F.: Uncertainty principles for Wigner-Ville distribution associated with the quaternion offset linear canonical transform. Circuits Syst. Signal Process. **42**(1), 385–404 (2023)
30. Kou, K.I., Ou, J., Morais, J.: Uncertainty principles associated with quaternionic linear canonical transforms. Math. Methods Appl. Sci. **39**(10), 2722–2736 (2016)
31. Cao, Y.J., Li, B.Z., Li, Y.-G., Chen, Y.H.: Logarithmic uncertainty relations for odd or even signals associate with Wigner-ville distribution. Circuits Syst. Signal Process. **35**(7), 2471–2486 (2016)
32. Bahri, M., Ashino, R.: Logarithmic uncertainty principle for quaternion linear canonical transform. In: 2016 International conference on wavelet analysis and pattern recognition (ICWAPR), pp. 140–145 (2016). IEEE
33. Kamel, B., Tefjeni, E.: Uncertainty principle for the two-sided quaternion windowed Fourier transform. Integral Transform. Spec. Funct. **30**(5), 362–382 (2019)

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